# What's so nice about the category of lenses?

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# COLLABORATORS

This talk is based on research from the Applied Category Theory 2020 Adjoint School together with:

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The goal of the project (and this talk) was to investigate the "nice" properties of the category $\mathfrak{Lens}$ of categories and (delta) lenses.

# OUTLINE OF THE TALK

**Background**

- What is a lens?
- Some basic examples

**Motivation**

- The problem with pullbacks
- A partial solution via discrete opfibrations

**Results**

- Products, equalisers, coproducts, extensivity
- Factorisation systems, epis, monos

**Conclusion**

- Ideas for future work
- Summary of the talk

# REVIEWING LENSES

A **lens** $(f, \varphi) : A \rightleftarrows B$ is a functor equipped with a **choice of lifts**,

$$\begin{tikzcd} A \arrow[d, shift left, "f"] & a \arrow[r, "\varphi(a,u)"] & a' \\ B \arrow[u, shift left, "\varphi"] & fa \arrow[r, "u"] & b \end{tikzcd}$$

which satisfies the axioms:

- $\varphi(a, 1_{fa}) = 1_a$
- $\varphi(a, v \circ u) = \varphi(a', v) \circ \varphi(a, u)$

- Every lens $(f, \varphi) : A \rightleftarrows B$ may be represented by a commutative diagram in $\mathcal{C}at$,

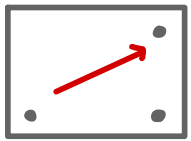
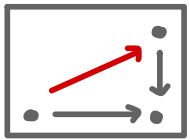
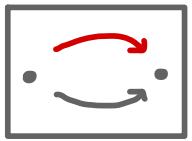
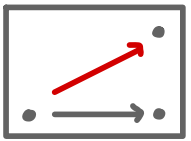
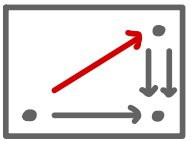
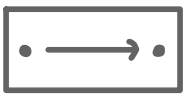
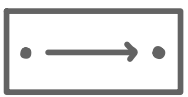
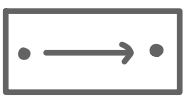
$$\begin{tikzcd} & \Lambda \arrow[dl, "\varphi"'] \arrow[dr, "\bar{\varphi}"] & \\ A \arrow[rr, "f"'] & & B \end{tikzcd}$$

where $\varphi$ is **bijective-on-objects** and $\bar{\varphi}$ is a **discrete opfibration**.

- The span $(\bar{\varphi}, \Lambda, \varphi) : B \nrightarrow A$ is called a **cofunctor**.

# BASIC EXAMPLES OF LENSES

Lenses without opcartesian lifts

| Discrete opfibration | Split opfibration | Bijective-on-objects | | |
|---|---|---|---|---|
|  |  |  |  |  |
| |  |  | |  |
| $\begin{tikzcd} & A \arrow[dr] & \\ A \arrow[ur, equal] \arrow[rr] & & B \end{tikzcd}$ | Chosen lifts $\varphi(a,u)$ are opcartesian | $\begin{tikzcd} & B \arrow[dl] \arrow[dr, equal] & \\ A \arrow[rr] & & B \end{tikzcd}$ | Existence of fillers fails | Uniqueness of fillers fails |

# THE CATEGORY OF LENSES

There is a category **Lens** whose objects are categories and whose morphisms are lenses with composition given by:

$$
\begin{tikzcd}
A \arrow[d, shift left, "f"] & a \arrow[r, "\varphi(a,\gamma(fa,u))"] \arrow[d, dotted, shift left] & a' \arrow[d, dotted, shift left] \\
B \arrow[u, shift left, "\varphi"] \arrow[d, shift left, "g"] & fa \arrow[r, "\gamma(fa,u)"] \arrow[d, dotted, shift left] & b \arrow[d, dotted, shift left] \\
C \arrow[u, shift left, "\gamma"] & gfa \arrow[r, "u"] & c
\end{tikzcd}
$$

$$
\begin{tikzcd}
 & & \bullet \arrow[dl] \arrow[dr] \arrow[d, phantom, "\cong"] & & \\
 & \Lambda \arrow[dl, "\varphi"'] \arrow[dr, "\bar{\varphi}"] & & \Omega \arrow[dl, "\gamma"'] \arrow[dr, "\bar{\gamma}"] & \\
A \arrow[rr, "f"'] & & B \arrow[rr, "g"'] & & C
\end{tikzcd}
$$

There is an identity-on-objects isofibration $\mathcal{L}ens \longrightarrow \mathcal{C}at$ which assigns a lens to it's underlying functor.

# THE PROBLEM WITH PULLBACKS

- A category $\mathcal{E}$ has pullbacks $\Leftrightarrow$ $\mathcal{E}/B$ has products for all $B \in \mathcal{E}$.
- The category $\mathcal{L}ens/B$ has a **monoidal product** given by:

$$\begin{tikzcd}
A \times_B \Omega \arrow[r, "1 \times \gamma"] \arrow[dr, "\bar{\pi}_A"'] & A \times_B C \arrow[d, phantom, "\lrcorner"] \arrow[dl, "\pi_A"] \arrow[dr, "\pi_C"'] & \Lambda \times_B C \arrow[l, "\varphi \times 1"'] \arrow[dl, "\bar{\pi}_C"] \\
& A \arrow[dr, "f"] & C \arrow[dl, "g"'] \\
\Lambda \arrow[uur, "\varphi"] \arrow[rr, "\bar{\varphi}"'] & & B & \Omega \arrow[ul, "\gamma"'] \arrow[ll, "\bar{\gamma}"]
\end{tikzcd}$$

- **Problem**: this monoidal structure is not cartesian in general!
- Is it possible that the universal property holds for certain inputs?

# DIGRESSION: SOME USEFUL LEMMAS

Consider a diagram in $\mathcal{C}at$ with $g$ a discrete opfibration:

$$\begin{tikzcd} A \arrow[rr, "f"] \arrow[dr, "g \circ f"'] & & B \arrow[dl, "g"] \\ & C & \end{tikzcd} \qquad (*)$$

Then:

(1) $g \circ f$ is a discrete opfibration $\Rightarrow$ $f$ is a discrete opfibration

(2) $g \circ f$ has a lens structure $\Rightarrow$ $f$ has a unique lens structure such that $(*)$ commutes in $\mathcal{L}ens$

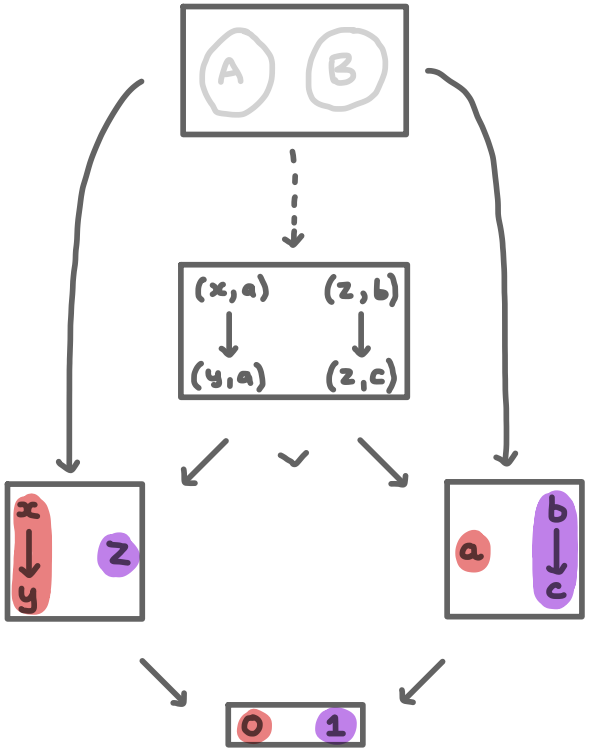
# PULLBACKS ALONG DISCRETE OPFIBRATIONS

**Proposition:** The category $\mathcal{L}ens$ admits all pullbacks along discrete opfibrations. Moreover, these pullbacks coincide with the "canonical" monoidal structure on $\mathcal{L}ens/B$.

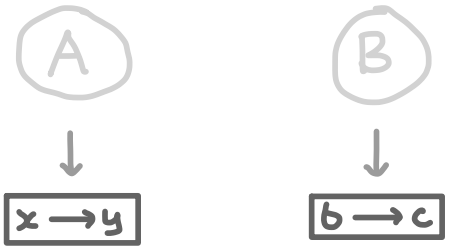
**Proof:** Apply the previous lemma to the triangle (*) to give the universal functor $Z \to A \times_B C$ a lens structure.

$$
\begin{tikzcd}
 & & Y \arrow[dl] \arrow[ddr, bend left] & & \\
 & Z \arrow[dr, dashed, "(*)"'] \arrow[ddr, bend left] \arrow[drr, bend left] & & & \\
 & & A \times_B C \arrow[dl] \arrow[dr] & & \\
X \arrow[uur] \arrow[dr] & & & C \arrow[dl] & \\
 & A \arrow[rr] & & B & \\
 & & \Lambda \arrow[ul] \arrow[ur] & &
\end{tikzcd}
$$

# OTHER EXAMPLES OF PULLBACKS




Any other commuting square of lenses into the cospan is equivalent to a pair of lenses from the connected components as follows:




Thus the universal property is satisfied.

**Conjecture:** If all products of fibres involve a discrete cat, the pullback exists.

# MONOIDAL STRUCTURE & PRODUCTS

The cartesian monoidal structure on $\mathcal{C}at$ induces a **semi-cartesian monoidal structure** on $\mathcal{L}ens$ with projections given by:

$$\begin{tikzcd}
 & A \times B_0 \arrow[dl, "\bar{\pi}_A"'] \arrow[dr, "1 \times i"] & & A_0 \times B \arrow[dl, "i \times 1"'] \arrow[dr, "\bar{\pi}_B"] & \\
A & & A \times B \arrow[ll, "\pi_A"] \arrow[rr, "\pi_B"'] & & B
\end{tikzcd}$$

Moreover, if $A$ or $B$ is a **discrete category**, then the above corresponds to the **cartesian product** in $\mathcal{L}ens$.

**Open question:** Are these the only products in $\mathcal{L}ens$?

# COPRODUCTS

**Proposition:** The canonical functor $\mathcal{L}ens \longrightarrow \mathcal{C}at$ **creates coproducts**.

**Proof (idea):** Recall that in $\mathcal{C}at$, the coproduct injection functors are injective-on-objects discrete opfibrations.

Given a pair of lenses $(f,\varphi): A \longrightarrow B$ and $(g,\gamma): C \longrightarrow B$ we have a unique lens from the coproduct in $\mathcal{C}at$ commuting with the injections given by:

$$\begin{tikzcd}
 & \Lambda + \Omega \arrow[dl, "\varphi+\gamma"'] \arrow[dr, "{[\bar{\varphi}, \bar{\gamma}]}"] & \\
A + C \arrow[rr, "{[f,g]}"'] & & B
\end{tikzcd}$$

# DISTRIBUTIVITY

- A monoidal category is **distributive** if the canonical map,

$$A \otimes B + A \otimes C \xrightarrow{[1 \otimes i_B,\ 1 \otimes i_C]} A \otimes (B + C)$$

is an isomorphism for all $A, B, C$.

- Since $\mathcal{C}at$ distributive (with respect to the cartesian monoidal structure), and $\mathcal{L}ens \longrightarrow \mathcal{C}at$ is a strong monoidal isofibration, we have that **$\mathcal{L}ens$ is distributive**.*

- Actually we can show that distributivity follows from an even stronger property of $\mathcal{L}ens$.

# EXTENSIVITY

- A category with coproducts is **extensive** if pullbacks along coproduct injections exist, and in any commutative diagram,

$$\begin{tikzcd}
X \arrow[r] \arrow[d] & Z \arrow[d] & Y \arrow[l] \arrow[d] \\
A \arrow[r, "\iota_A"'] & A+B & B \arrow[l, "\iota_B"]
\end{tikzcd}$$

the two squares are pullbacks $\Leftrightarrow$ the top row is a coproduct.

- **Proposition**: $\mathcal{L}ens$ is an extensive category.

# EQUALISERS

- **Lens admits all equalisers**, however unlike pullbacks and products, they are not always preserved by $\mathcal{L}ens \to \mathcal{C}at$.

- We may construct equalisers in $\mathcal{L}ens$ through taking the **largest subobject*** of the equaliser in $\mathcal{C}at$ which admits a lens structure and equalises the diagram in $\mathcal{L}ens$.

$$
\begin{tikzcd}
 & & d \arrow[d] \\
a \arrow[r] \arrow[urr] \arrow[rr, bend right] & b \arrow[r] & c \\
 & & \\
x \arrow[r] \arrow[rr, bend right] & y \arrow[r] & z
\end{tikzcd}
\qquad
\begin{tikzcd}
 & d \arrow[dr] & \\
a \arrow[r] \arrow[rr, bend right] & b \arrow[r] & c \\
 & & \\
x \arrow[r] \arrow[rr, bend right] & y \arrow[r] & z
\end{tikzcd}
\quad \rightsquigarrow \quad
\begin{tikzcd}
 & b \arrow[r] & c \\
a \arrow[ur] \arrow[urr] & & \\
 & & \\
 & b \arrow[r] & c \\
a \arrow[ur] \arrow[urr] \arrow[r] & d \arrow[ur] & \\
 & &
\end{tikzcd}
$$

# MONOMORPHISMS

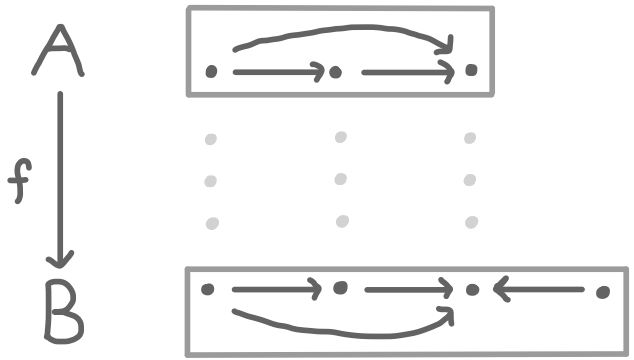
A lens $(f,\varphi): A \rightleftarrows B$ is a **monomorphism** if any of the following equivalent conditions hold:

(1) $(f,\varphi)$ is an **injective-on-objects discrete opfibration**;

(2) $(f,\varphi)$ is a **fully faithful discrete opfibration**;

(3) $f$ is a **monomorphism** in $\mathcal{C}at$.

**Conjecture:** These conditions are necessary and sufficient.



# EPIMORPHISMS

A lens $(f,\varphi): A \rightleftarrows B$ is an **epimorphism** if any of the following equivalent conditions hold:

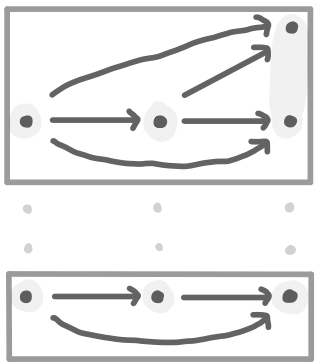
(1) $(f,\varphi)$ is **surjective-on-objects**;

(2) $(f,\varphi)$ is **surjective-on-morphisms**.

**Conjecture:** These conditions are necessary and sufficient.

**Corollary (?):** In $\mathcal{L}ens$, epi + mono $\Longleftrightarrow$ isomorphism.

$$\begin{tikzcd} A \arrow[d, shift right, "f"'] \\ B \arrow[u, shift right, "\varphi"'] \end{tikzcd}$$



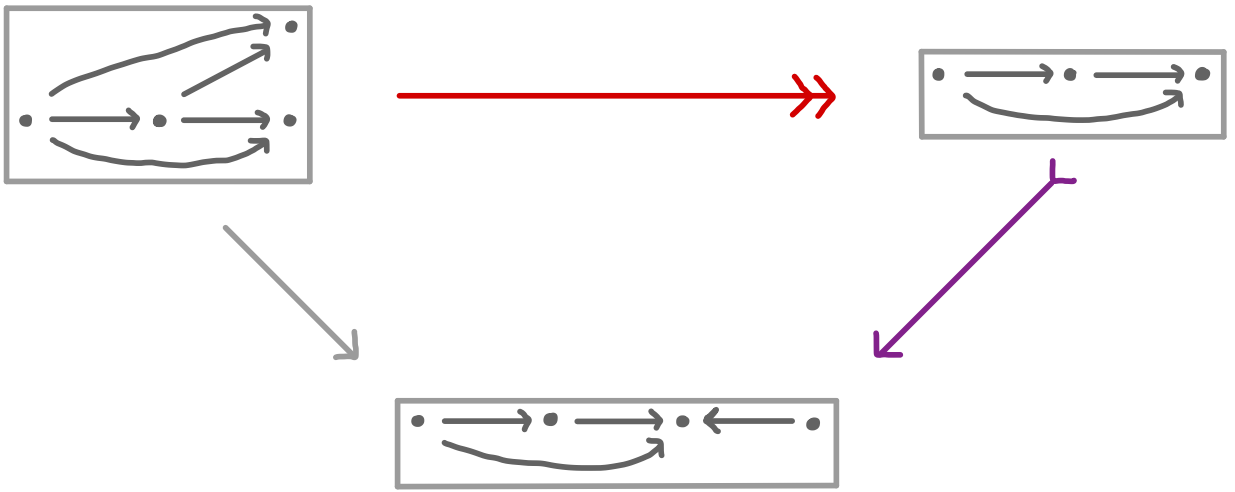
# PROPER FACTORISATION SYSTEM

The category $\mathcal{L}ens$ has an orthogonal factorisation system with:

$\mathcal{E}$ = surjective-on-objects lenses $\subseteq \mathrm{Epi}(\mathcal{L}ens)$

$\mathcal{M}$ = injective-on-objects discrete opfibrations $\subseteq \mathrm{Mono}(\mathcal{L}ens)$

This corresponds to the (surj-on-ob, inj-on-ob f.f.) OFS via $\mathcal{L}ens \to \mathcal{C}at$.



# SPLIT IDEMPOTENTS

- An idempotent splits if the following parallel pair has an equaliser (or coequaliser):

$$\begin{tikzcd} Q \arrow[r, dashed] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "1_A"'] & A \arrow[r, dashed] & Q \end{tikzcd}$$

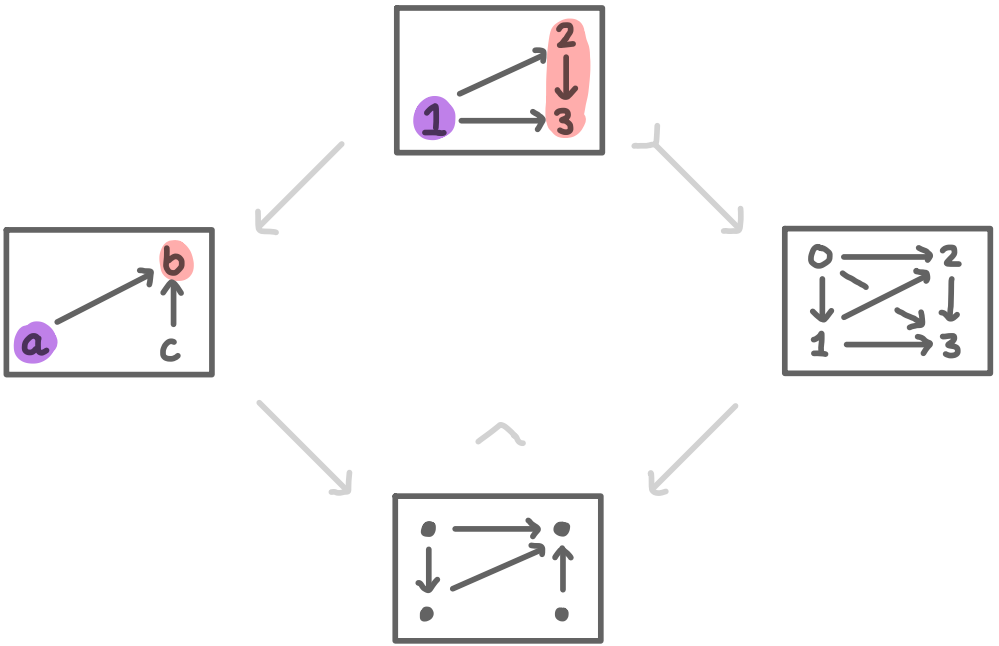
- In $\mathcal{L}ens$, all idempotents split and they may be obtained by the (epi, mono) factorisation:

$$\begin{tikzcd} A \arrow[rr, "f"] \arrow[dr, "e"'] & & A \\ & Q \arrow[ur, tail, "m"'] & \end{tikzcd}$$

- These give simple examples of coequalisers – are there others?

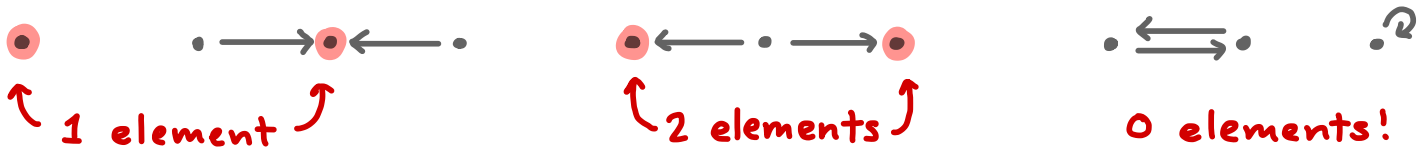
# PUSHOUTS & ADHESIVITY?

- Like coequalisers, it seems difficult to construct pushouts in Lens, but do we at least have them along monos?



# ELEMENTS & SINKS

- For categories and functors we have $\mathcal{C}at(1, A) \cong A$, but this statement is not true in $\mathcal{L}ens$!




- The set of elements $\mathcal{L}ens(1, A)$ provides an invariant for the category which measures the amount of **sinks** in $A$, the set of elements $a \in A$ such that $\sum_{x \in A} A(a, x) = 1$.
- The set of sinks is a **right adjoint** to the the discrete category functor:

$$\mathcal{L}ens(\mathrm{disc}(A), B) \cong \mathcal{S}et(A, \mathrm{sink}(B))$$

# MONOIDAL CLOSEDNESS

- The hom-sets in $\mathcal{L}ens$ may be given a category structure:

$$
\begin{tikzcd}
\wedge \arrow[dr, "\varphi"] \arrow[dd, "t"'] & & \\
\arrow[rr, phantom, "\Downarrow\tau"] & A \arrow[r, bend left, "f"] \arrow[r, bend right, "g"'] \arrow[r, phantom, "\Downarrow\sigma"] & B \\
\Omega \arrow[ur, "\gamma"'] & &
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
\wedge \arrow[dr, "\bar{\varphi}"] \arrow[dd, "t"'] & \\
& B \\
\Omega \arrow[ur, "\bar{\gamma}"'] &
\end{tikzcd}
$$

$$A \otimes (-) \cong$$

- One might suspect that $\mathcal{L}ens(A,-)$ is right adjoint to $(-)\otimes A$.
- However this is not true! Consider $A = \{\bullet \rightleftarrows \bullet\}$ and so

$$\frac{1 \otimes A = A \longrightarrow A}{A \longrightarrow \mathcal{L}ens(1,A) = \emptyset}$$

Contradiction!

- Is it possible that $(\mathcal{L}ens, \otimes, 1)$ is monoidal closed?

# SUMMARY & FUTURE WORK

- In this talk, we have seen that $\mathcal{L}ens$ a many "nice" aspects including:
  - pullbacks along discrete opfibs.
  - semi-cartesian monoidal structure
  - coproducts
  - distributivity and extensivity
  - equalisers
  - proper factorisation system
  - sufficient conditions for epi/mono

- What are categories internal to $\mathcal{L}ens$ with source map a discrete opfibration?
- When $\mathcal{V} = \mathcal{L}ens$, is the bicategory $\mathcal{V}$-Mat interesting?
- What about $\mathcal{L}ens$-enriched cats?
- Does $\mathcal{L}ens$ admit other nice factorisation systems or monoidal structures?
- Is $\mathcal{L}ens$ a "nice" 2-category?